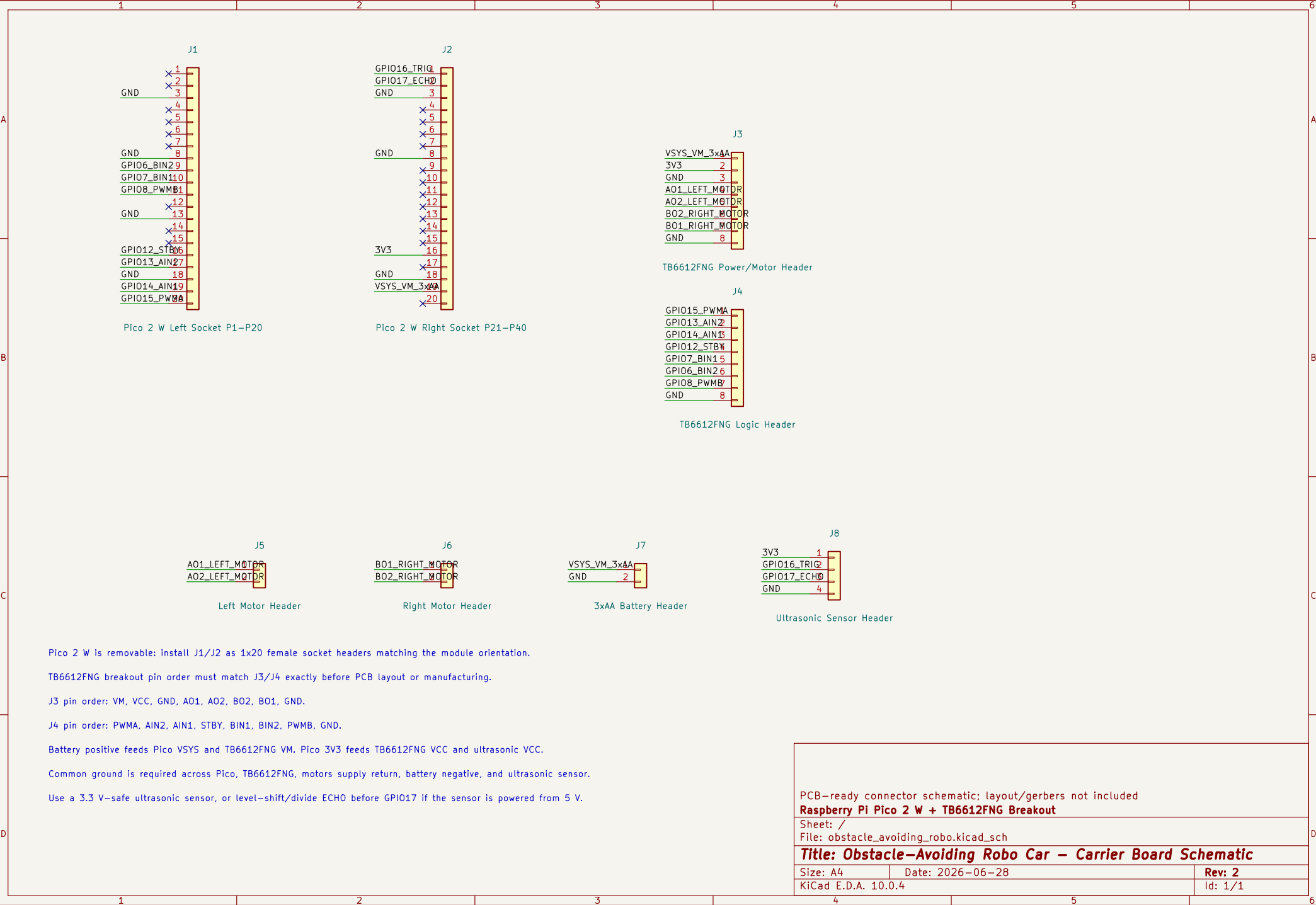




Pico 2 W is removable: install J1/J2 as 1x20 female socket headers matching the module orientation.

TB6612FNG breakout pin order must match J3/J4 exactly before PCB layout or manufacturing.

J3 pin order: VM, VCC, GND, AO1, AO2, BO2, BO1, GND.

J4 pin order: PWMA, AIN2, AIN1, STBY, BIN1, BIN2, PWMB, GND.

Battery positive feeds Pico VSYS and TB6612FNG VM. Pico 3V3 feeds TB6612FNG VCC and ultrasonic VCC.

Common ground is required across Pico, TB6612FNG, motors supply return, battery negative, and ultrasonic sensor.

Use a 3.3 V–safe ultrasonic sensor, or level–shift/divide ECHO before GPIO17 if the sensor is powered from 5 V.

PCB–ready connector schematic; layout/gerbers not included		
Raspberry Pi Pico 2 W + TB6612FNG Breakout		
Sheet: /		
File: obstacle_avoiding_robo.kicad_sch		
Title: Obstacle–Avoiding Robo Car – Carrier Board Schematic		
Size: A4	Date: 2026–06–28	Rev: 2
KiCad E.D.A. 10.0.4	Id: 1/1	